

NAND TO TETRIS ASSIGNMENT

Assessment Number: 1

Course: B.Tech CSE AI and ML

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Section: F

Subject: NAND to Tetris

Question 1 :

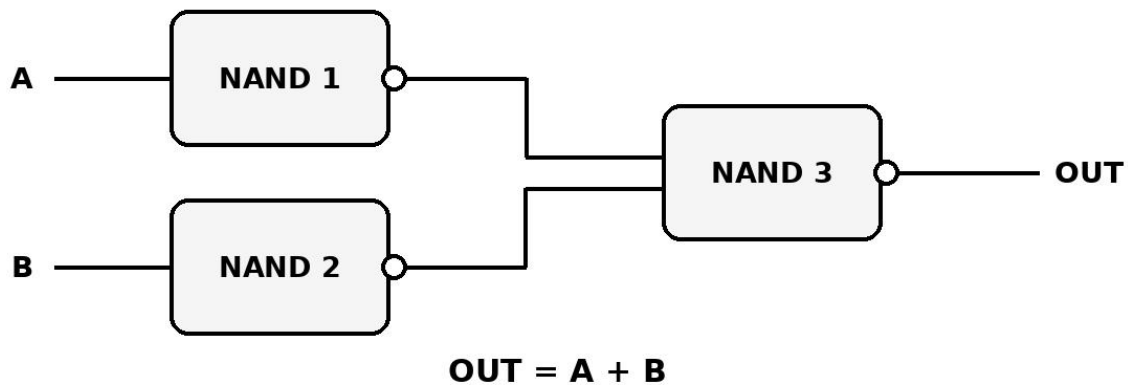
Implement NOT, AND, and OR gates using only NAND gates in HDL. Draw the logic diagram for each and write the Boolean expression. Run the provided test scripts and capture the simulation output.

Answer:

1. NOT Gate

➤ Logic Diagram

OR Gate Using NAND Gates

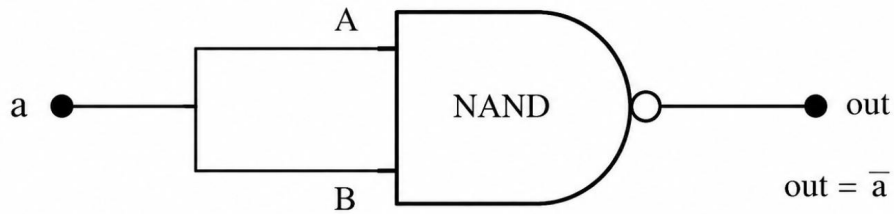


Boolean Expression: $NOT(a) = a \text{ NAND } a = a'$

➤ Truth Table of NOT Gate

NOT Gate Using NAND Gate

Logic Diagram:



Both inputs of the NAND gate are connected to the same input 'a'.
Therefore, the output is the complement of a.

Boolean Expression:

$$\text{NOT}(a) = a \text{ NAND } a = \bar{a}$$

➤ HDL Dashboard

```
Test [Icons] Steps: 7 Outputs: 5
Slow [Slider] Fast

Test Script | Compare File | Output File | Diff Table
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/And.tst
5
6 load And.hdl,
7 compare-to And.cmp,
8 output-list a b out;
9
10 set a 0,
11 set b 0,
12 eval,
13 output;
14
15 set a 0,
16 set b 1,
17 eval,
18 output;
19
20 set a 1,
21 set b 0,
22 eval,
23 output;
24
25 set a 1,
26 set b 1,
27 eval,
28 output;
```

➤ Test Script

Chip And Eval Reset Clock: 0

Input pins

a 1

b 1

Output pins

out 1

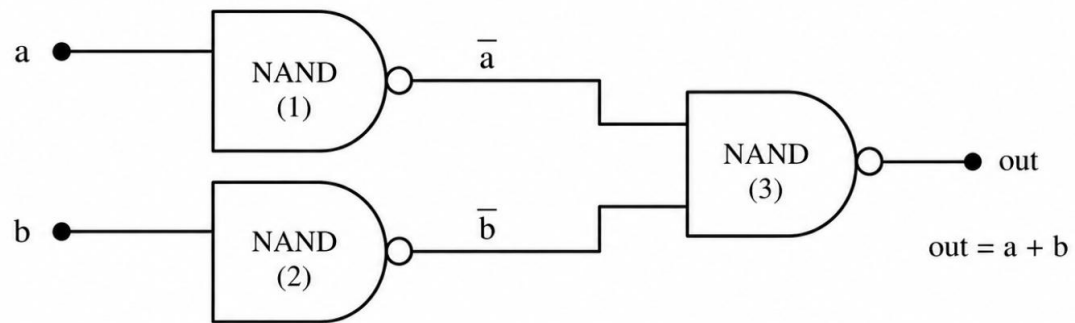
Internal pins

nandOut 0

➤ Pins

Test				Steps: 7 Outputs: 5			
Slow				Fast			
Test Script		Compare File		Output File		Diff Table	
	a	b	out				
	0	0	0				
	0	1	0				
	1	0	0				
	1	1	1				

➤ Comparison Table

Logic Diagram:

The first NAND gate inverts 'a' to $\bar{a}$.
The second NAND gate inverts 'b' to $\bar{b}$.
The third NAND gate computes the NAND of $\bar{a}$ and $\bar{b}$.
By De Morgan's law, $\overline{\bar{a} \cdot \bar{b}} = a + b$.

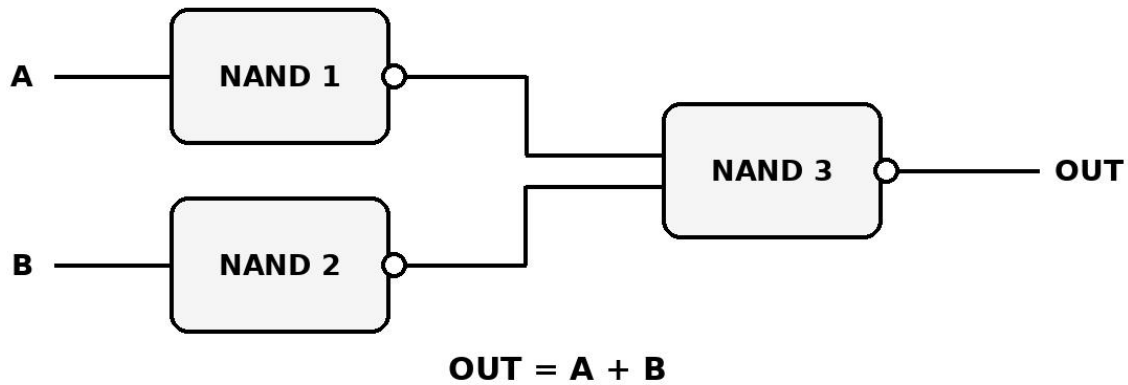
Boolean Expression:

$$\text{OR}(a, b) = \overline{\bar{a} \cdot \bar{b}} = a + b$$

2. AND Gate

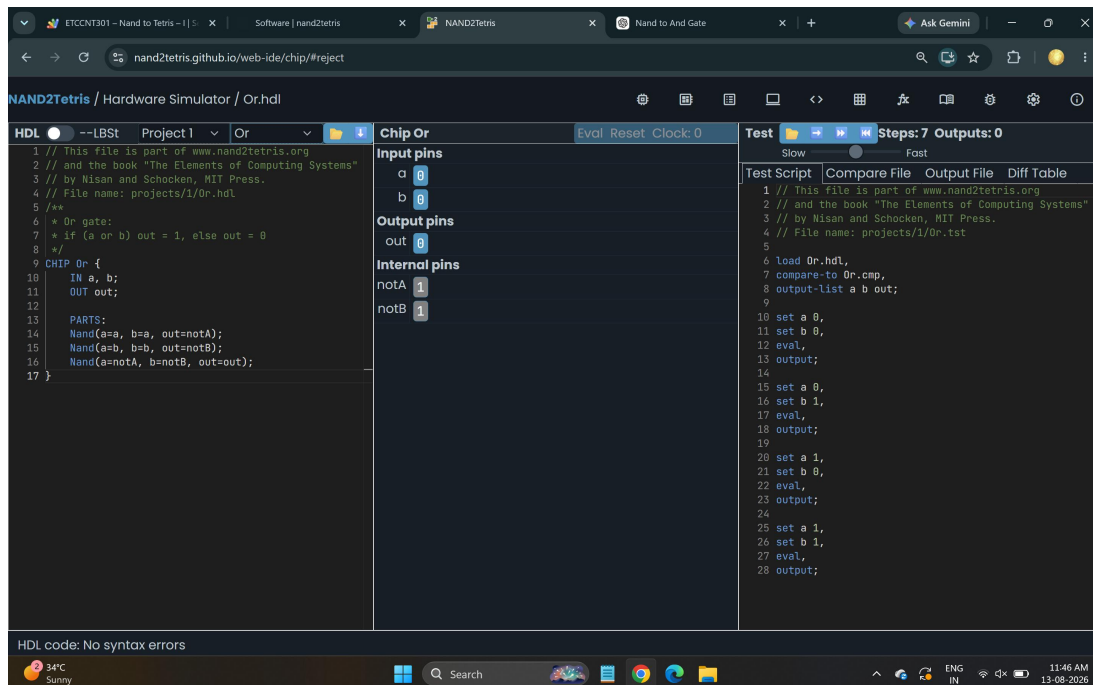
➤ Logic Diagram

OR Gate Using NAND Gates



Boolean Expression: $AND(a,b) = a \cdot b$

➤ Truth Table



➤ HDL Dashboard

Test     **Steps: 7 Outputs: 0**

Slow  Fast

Test Script	Compare File	Output File	Diff Table
<pre>1 // This file is part of www.nand2tetris.org 2 // and the book "The Elements of Computing Systems" 3 // by Nisan and Schocken, MIT Press. 4 // File name: projects/1/0r.tst 5 6 load 0r.hdl, 7 compare-to 0r.cmp, 8 output-list a b out; 9 10 set a 0, 11 set b 0, 12 eval, 13 output; 14 15 set a 0, 16 set b 1, 17 eval, 18 output; 19 20 set a 1, 21 set b 0, 22 eval, 23 output; 24 25 set a 1, 26 set b 1, 27 eval, 28 output;</pre>			

➤ Test Script

Chip Or

Eval Reset Clock: 0

Input pins

a

0

b

0

Output pins

out

0

Internal pins

notA

1

notB

1

➤ Pins

Test

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Steps: 7 Outputs: 5

Slow

⬤

Fast

Test Script	Compare File	Output File	Diff Table															
<table> <thead> <tr> <th>a</th> <th>b</th> <th>out</th> </tr> </thead> <tbody> <tr> <td>0</td> <td>0</td> <td>0</td> </tr> <tr> <td>0</td> <td>1</td> <td>0</td> </tr> <tr> <td>1</td> <td>0</td> <td>0</td> </tr> <tr> <td>1</td> <td>1</td> <td>1</td> </tr> </tbody> </table>	a	b	out	0	0	0	0	1	0	1	0	0	1	1	1			
a	b	out																
0	0	0																
0	1	0																
1	0	0																
1	1	1																

➤ Comparison Table

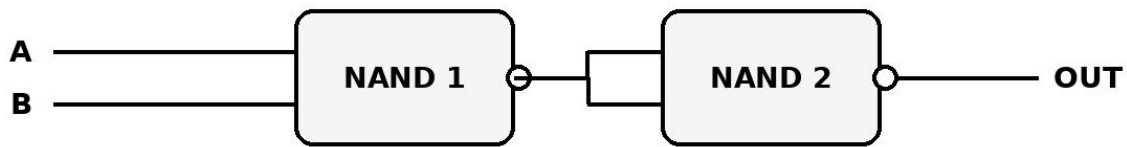
Truth Table

A (Input)	B (Input)	$Y = (\overline{AB}) = \overline{A}$
0	0	1
1	1	0

3. OR Gate

➤ Logic Diagram

AND Gate Using NAND Gates



$$\text{OUT} = A \cdot B$$

Boolean Expression: $\text{OR}(a,b) = a + b$

➤ Truth Table

```
Test [Folder] [Next] [Run] [Previous] Steps: 5 Outputs: 3
Slow [Slider] Fast

Test Script | Compare File | Output File | Diff Table
1 This file is part of www.nand2tetris.org
2 and the book "The Elements of Computing Systems"
3 by Nisan and Schocken, MIT Press.
4 File name: projects/1/Not.tst
5
6 d Not.hdl,
7 pare-to Not.cmp,
8 put-list in out;
9
10 in 0,
11 l,
12 put;
13
14 in 1,
15 l,
16 put;
```

➤ HDL Dashboard

Chip Not

Eval Reset Clock: 0

Input pins

in 1

Output pins

out 0

➤ Test Script

Test

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Steps: 5 Outputs: 3

Slow

Fast

Test Script	Compare File	Output File	Diff Table
<pre> in out 0 1 1 0 </pre>			

➤ Pins

AND Gate Using NAND Gates

Logic Diagram:

The first NAND gate produces the complement of $(a \cdot b)$, i.e., $x = \overline{a \cdot b}$.
 The second NAND gate inverts x to give the final output: $out = \overline{x} = a \cdot b$.

Boolean Expression:

$$AND(a, b) = \overline{\overline{a \cdot b}} = a \cdot b$$

➤ Comparison Table

Input A	Input B	Output ($A \cdot B$)
0	0	0
0	1	0
1	0	0
1	1	1

Conclusion

The NAND gate can be used as a universal logic gate to construct NOT, AND, and OR operations.

